

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas, Fayetteville, AR

Aug 2024 – May 2026

Master of Science in Computer Engineering

Bachelor of Engineering in Electronics and Communication, India

Aug 2019 – May 2023

Sri Ramakrishna Engineering College

TECHNICAL SKILLS

Telecommunications: Wireless Communication, RF Systems, Antenna Design, Antenna Simulation, 5G, IoT, MQTT, Wireless Sensor Systems, Communication Interfaces, Serial Communication

RF & Microwave: Ansys HFSS, Electromagnetic Simulation, SIW (Substrate Integrated Waveguide), Deep-Well Cavity, Antenna Optimization, RF Design, Antenna Fabrication, 5G/Biosensing Antennas

Communication Protocols: UART, SPI, I2C, MQTT, AXI, Serial Communication, Sensor Interfaces, Inter-Module Communication

Embedded & Digital Systems: Embedded C, C, C++, C#, Python, Microcontrollers, Microprocessors, Firmware Development, FreeRTOS, Real-Time Systems, FPGA, SoC, MicroBlaze, VHDL, Verilog, Assembly, AXI-Based IP Cores

Signal & Control Systems: Sensors, ADC, GPIO, PWM, PID Control, Feedback Control, Servo Control, Motor Control, Data Acquisition, Real-Time Control

Hardware & Simulation Tools: Vivado, Vitis, MPLAB IDE, Proteus, Ansys HFSS, LabVIEW, NI TestStand, MAVIS

Development Tools: Git, Bitbucket, Jira, Agile, Scrum, SDLC, Object-Oriented Programming

PROJECTS

Biosensing Antenna Design

Ansys HFSS, RF Simulation, Antenna Design

- Designed and analyzed an antenna for 5G and biosensing applications using Ansys HFSS and electromagnetic simulation techniques
- Calculated and optimized unit-cell dimensions and antenna geometry to improve the desired electromagnetic response
- Performed iterative simulations and parametric optimization to evaluate antenna characteristics and refine the design
- Applied RF, microwave, and electromagnetic design principles for antenna modeling, performance analysis, and optimization

SIW & Deep-Well Cavity Antenna

Ansys HFSS, SIW, RF/Microwave

- Designed and simulated antenna structures incorporating Substrate Integrated Waveguide (SIW) and deep-well cavity techniques
- Performed full-wave electromagnetic simulations and parametric optimization using Ansys HFSS
- Analyzed the impact of antenna dimensions and structural parameters on simulated electromagnetic performance
- Supported antenna fabrication and physical implementation, gaining experience across RF design, simulation, optimization, and prototyping

PROFESSIONAL EXPERIENCE

Project Engineer, Soliton Technologies, India

June 2023 – June 2024

- Developed a C#/.NET WPF desktop application for Dräger integrating C#, Python, and LabVIEW-based test automation, improving test workflow efficiency, user onboarding, and automated test coverage for medical-device engineering
- Developed customized automated testing solutions using MAVIS, an object-oriented test automation framework built on NI LabVIEW and Python, supporting verification and validation (V&V) activities for medical-device applications
- Served as the sole developer on a Medtronic project, automating approximately 94% of manual test cases while supporting quality and compliance requirements in a medical-device development environment
- Integrated software components and test automation workflows across C#, Python, LabVIEW, and TestStand, demonstrating hardware/software integration and automated validation capabilities

Engineering Intern, Soliton Technologies, India

July 2022 – May 2023

- Received intensive hands-on training in LabVIEW, Object-Oriented Programming, TestStand, and automated testing, developing applications for verification and validation workflows
- Worked with MAVIS, a LabVIEW/Python-based test automation framework, to streamline automated testing processes for medical-device engineering projects

Intern, National Institute of Technology, India

Sep 2021- Oct 2021

- Conducted research on antenna design incorporating Substrate Integrated Waveguide (SIW) and deep-well cavity structures to enhance antenna performance
- Designed and modeled antenna structures using Ansys HFSS for electromagnetic simulation and performance analysis
- Performed simulation-based optimization of antenna geometry and structural parameters to improve the desired electromagnetic characteristics
- Supported antenna fabrication and physical implementation, gaining hands-on exposure to the RF design, simulation, optimization, and prototyping workflow